EXECUTIVE PITCH | PROBLEM DEFINITION & BIOPHYSICAL FRONTIER

Engineering Nature's Electrical Wiring: Predicting Metalloprotein Redox Potentials

Why Iron-Sulfur ([Fe-S]) cluster redox potential (Em) prediction is a multi-million dollar biophysical bottleneck

1. WHAT WE DO & WHY IT MATTERS

- **Nature's Universal Bio-Batteries:**

  Fe-S proteins power respiration (Complex I), photosynthesis, and nitrogenase.

- **Hyper-Sensitivity (60 mV Rule):**

  A ±60 mV shift alters electron transfer rates by 10x, halting cellular metabolism.

- **The High-Value Mission:**

  Direct 3D structure-to-voltage prediction unlocks rational de novo metalloenzyme design.

- **The Current Bottleneck:**

  Lab purification costs $1,000s/mutant; quantum DFT takes days and ignores

2. WHY IT IS AN EXTREMELY HARD PROBLEM

- **The 1,000 mV Paradox:**

  Identical [4Fe-4S] cubanes span -700 mV (Fd) to +500 mV (HiPIP).

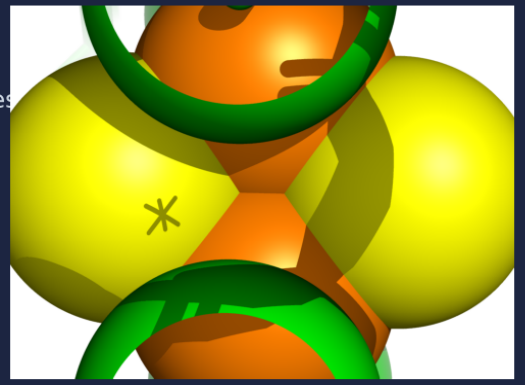
- **Sub-Angstrom Quantum Sensitivity:**

  A single N-H···S hydrogen bond shifts Em by +60 to +100 mV.

- **Solvent & pH Coupling:**

  Dielectric shielding and pH titration control active site thermodynamics.

3. QUANTUM ACTIVE SITE: [4Fe-4S] CLUSTER MICROENVIRONMENT



▸ **Inorganic Cubane Core**

[Fe4S4] redox center

▸ **Cysteine Coordination**

4 Cys-Sγ ligands

▸ **Outer Dielectric Shell**

Polarity & pH titration

METHODOLOGY & ARCHITECTURAL INNOVATION | FiLM-RGCN

Overcoming Broken Baselines with Physics-Conditioned Graph Message Passing

Diagnosing literature feature collisions and establishing a continuous 5-relational graph architecture

PRIOR LITERATURE & HIDDEN PITFALLS

✖ Published Baseline (JCIM 2025):

XGBoost on raw tables yielded ~143-165 mV error on full data.

✖ Hidden Pipeline Bug Discovered:

Dictionary key overwrites generated hundreds of corrupted NaN columns.

✖ Vanilla GNNs Fail:

Signal dilution across 300+ residues washes out the active site.

OUR NOVEL SOLUTION: FiLM-RGCN

✔ Curated 96 Biophysical Descriptors:

Non-redundant curated features drop baseline to 48.26 mV.

✔ 5-Relational Edge Topology:

Explicitly models backbone, Fe-S, H-bonds, hydrophobic, & cluster edges.

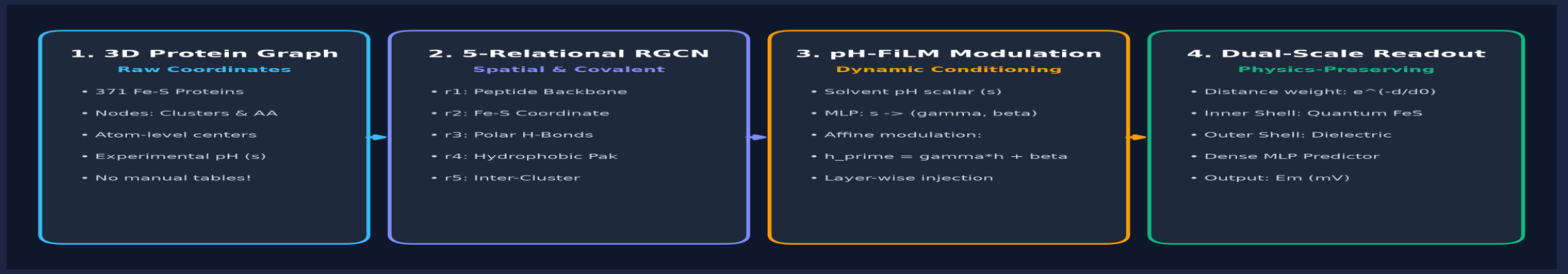
✔ Layer-Wise pH-FiLM Conditioning:

Injects experimental pH as affine modulation (h' = γ(pH)⊙h + β(pH)).

✔ Dual-Scale Exponential Readout:

Continuous physics decay w(d) = exp(-d/d0) preserves inner quantum shell.

END-TO-END FiLM-RGCN DEEP LEARNING PIPELINE

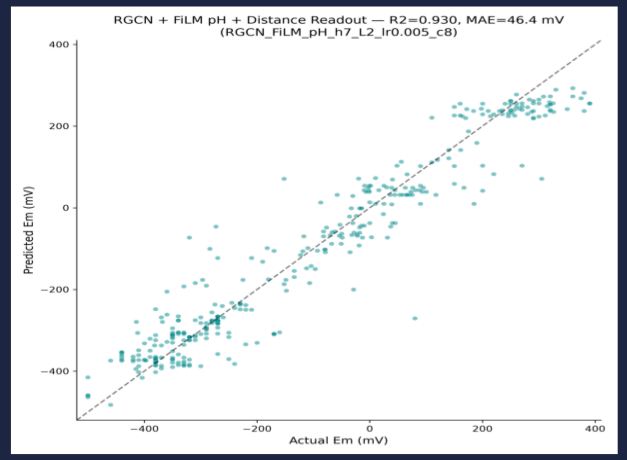


BREAKTHROUGH RESULTS & REAL-WORLD APPLICATION | IMPACT PITCH

Unprecedented Accuracy, Mechanistic Explainability & In Silico Design

Setting the new state-of-the-art and delivering actionable screening for de novo protein engineering

1. BENCHMARK TRIUMPH



- **Champion MAE: 46.35 mV**

  R^2 = 0.929 | Pearson r = 0.965

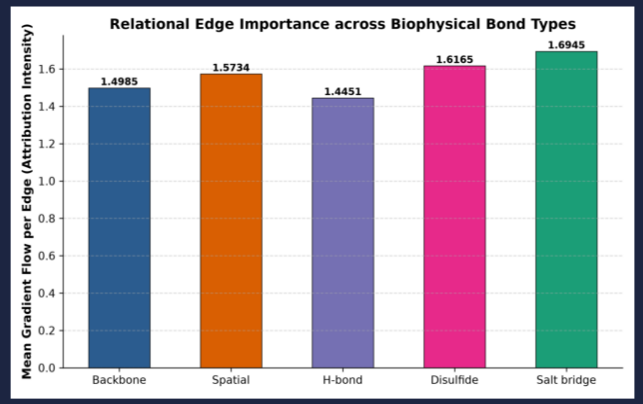
- **>70% Error Reduction**

  Slashes 164.8 mV baseline error

- **Universal Parity Agreement**

  Full -700 mV to +500 mV range

2. EXPLAINABLE AI (XAI)



▸ **N-H···S Hydrogen Bonds:**

Top driver: +60 to +100 mV per bond

▸ **Solvent Dielectric (SASA):**

Water exposure depresses Em

▸ **pH Nernstian Titration:**

Dynamic PCET proton-coupled shifts

▸ **Model Learns Real Biophysics**

3. DE NOVO DESIGN IMPACT

★ **De Novo Protein Design:**

Screen millions of RFdiffusion backbones in seconds for desired voltages.

★ **Zero Wet-Lab Wastage:**

Filter out 95% of non-viable mutants prior to gene synthesis.

★ **Green Tech & Bio-Hydrogen:**

Tune hydrogenases to match electrode potentials for solar fuels.

★ **Metabolic Pathway Balance:**

Eliminate redox bottlenecks in engineered bacterial synthesis.

★ **Instant Laptop Inference:**

<15 ms per protein (10,000x faster than supercomputer DFT).